

Name _____ Date _____ Period _____

Graphing $f(x) = |x|$ and Key Attributes

Algebra II Honors | Unit 2 | Day 1 | TEK A2.2.A

Objective: I can graph $f(x) = |x|$ from its piecewise definition, derive its key attributes algebraically, and compare its behavior to a linear function.

Vocabulary

Piecewise function — a function defined by _____ on different parts of the domain.

Absolute value, formally — $|x| = x$ if $x \geq 0$, and $|x| = \underline{\hspace{1cm}}$ if $x < 0$.

Even function — $f(-x) = f(x)$ for every x in the domain; the graph is symmetric about the _____.

Vertex — where the two pieces meet: _____ for $f(x) = |x|$.

Part 1 — The Piecewise Definition

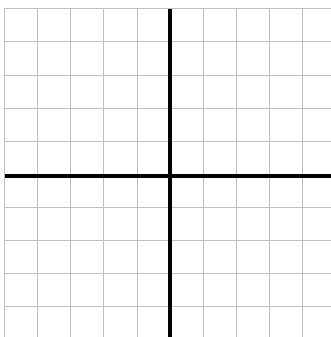
$$f(x) = |x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0 \end{cases}$$

Evaluate using the correct piece.

$$f(-4) = \underline{\hspace{1cm}} \text{ (uses } -x) \quad f(6) = \underline{\hspace{1cm}} \text{ (uses } x) \quad f(0) = \underline{\hspace{1cm}} \quad f(-0.5) = \underline{\hspace{1cm}} \text{ (uses } -x)$$

Part 2 — Graph Each Piece

Graph $y = x$ for $x \geq 0$ and $y = -x$ for $x < 0$ on the SAME grid - two rays meeting at the vertex. Each square is 1 unit, both axes run -5 to 5.



Part 3 — Key Attributes, Derived From the Definition

Domain: _____ Vertex: _____

Derive the range from the two pieces, instead of just reading it off the graph:

For $x \geq 0$, the rule is $y = x$, so y takes every value _____

For $x < 0$, the rule is $y = -x$, so y takes every value _____ (since x is negative, $-x$ is positive)

Combining both pieces, the range is _____

Now show algebraically that $f(x) = |x|$ is an even function:

$f(-x) = |-x| = \underline{\hspace{1cm}} \rightarrow$ confirmed even, so the graph is symmetric about the _____

Now derive the x - and y -intercepts from the definition instead of reading them off the graph:

x -piece: $x = 0 \rightarrow x = \underline{\hspace{1cm}}$ $-x$ -piece: $-x = 0 \rightarrow x = \underline{\hspace{1cm}}$ (both agree)

So the only x -intercept is _____, which is also the y -intercept, since $f(0) = \underline{\hspace{1cm}}$

Part 4 — Compare and Explain

1. Explain why $f(x) = |x|$ has a minimum value but a line $y = mx + b$ ($m \neq 0$) does not.

Sample: _____

2. For which values of x does $|x| = x$ (the two pieces agree)? _____
3. For which values of x does $|x| = -x$? _____ (at $x = 0$ both pieces give the same value, 0)
4. Is $f(x) = |x|$ a function? Justify using the vertical line test.

Yes _____

Part 5 — Real-World: Distance, Justified

A hiker's position is x miles from a trailhead at 0 (negative = west, positive = east). Distance from the trailhead is $d(x) = |x|$.

Using the piecewise definition, explain why $d(x)$ is never negative, no matter which direction the hiker walks.

Sample: _____

Part 6 — Going Further

1. For which value(s) of x does $|x| = 5$? _____

Explain in one sentence why there are two answers.

Sample: _____

2. For which value(s) of x does $|x| = -3$? _____

Explain why, using the range you derived in Part 3.

Sample: _____

3. Could $|x|$ ever equal a number strictly between -1 and 0, like -0.5? Explain.

No _____

Exit Ticket

Justify algebraically that $f(x) = |x|$ is symmetric about the y -axis, then give its domain, range, and vertex.

Domain _____ Range _____ Vertex _____

Transformations of $f(x) = |x|$, Part I

Algebra II Honors | Unit 2 | Day 2 | TEK A2.6.C

Objective: I can describe, graph, and justify the effect of a and b on $f(x) = a|bx|$, including why some transformations produce no visible change.

Vocabulary

Vertical stretch/compression — $af(x)$; $|a| > 1$ stretches away from the x-axis, $0 < |a| < 1$ compresses toward it.

Horizontal stretch/compression — $f(bx)$; $|b| > 1$ compresses toward the y-axis, $0 < |b| < 1$ stretches away from it.

Invariant point — a point that does NOT move under a transformation. For $f(x) = a|bx|$, the _____ is always invariant.

Part 1 — The Effect of a and b, Combined

Function	a	b	Effect
$f(x) = 2 x $	2	1	_____
$f(x) = (1/2) x $	1/2	1	_____
$f(x) = 2x $	1	2	_____
$f(x) = -2 3x $	-2	3	_____

Part 2 — Why the Vertex Never Moves

Show algebraically that (0, 0) is on the graph of $f(x) = a|bx|$ no matter what a and b are:

$f(0) = a|b \cdot 0| = a|0| = a \cdot \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \rightarrow$ the vertex (0, 0) is _____ under every a and b

Part 3 — Why Reflecting Over the y-axis Does Nothing

$f(x) = |-x|$ should reflect the parent over the y-axis ($b = -1$). Prove algebraically that it produces the SAME graph as $f(x) = |x|$:

$|-x| = |-1| \cdot |x| = \underline{\hspace{2cm}} \rightarrow$ identical to the parent - this is the same even-function fact from Day 1 ($f(-x) = f(x)$)

Part 4 — Which Graphs Are Really the Same?

For each pair, determine whether the two functions produce the SAME graph. Justify using the properties of absolute value.

1. $f(x) = 3|x|$ and $f(x) = |3x|$ _____ since $|3x| = |3||x| = 3|x|$

2. $f(x) = -2|x|$ and $f(x) = 2|-x|$ _____ $-2|x|$ opens downward; $2|-x| = 2|x|$ opens upward

3. $f(x) = (1/2)|4x|$ and $f(x) = 2|x|$ _____ since $(1/2)|4x| = (1/2)(4)|x| = 2|x|$

4. $f(x) = |6x|$ and $f(x) = 6|x|$ _____ since $|6x| = |6||x| = 6|x|$

Part 5 — Going Further: Factoring Out a Constant

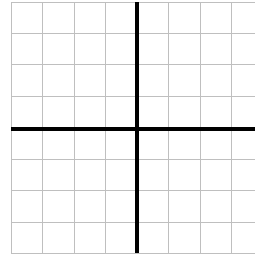
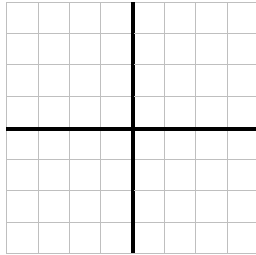
Tomorrow's lesson adds shifts: $f(x) = a|x - c| + d$. Some functions LOOK like they have a b-value inside, but a constant can be factored out to reveal the real a. Try it:

$f(x) = 2|3x - 6| = 2|3(x - 2)| = 2 \cdot \underline{\hspace{1cm}} \cdot |x - 2| = \underline{\hspace{1cm}}|x - 2| \rightarrow$ the REAL a-value is _____

$$f(x) = 4|2(x+4)| = \underline{\hspace{2cm}} |x+4| \rightarrow a = \underline{\hspace{2cm}} \text{ vertex x-coordinate } \underline{\hspace{2cm}}$$

Graph both on their own grid. Each square is 1 unit; both axes run -5 to 5.

B. $f(x) = (1/2)|x|$



1. Factor $f(x) = 3|6x + 18|$ into a $a|x - c|$ form and give the real a-value. _____

2. Order from flattest to steepest: $|x|$, $3|x|$, $(1/4)|x|$, $-2|x|$. _____

3. True or False: $f(x) = |-5x|$ and $f(x) = |5x|$ produce different graphs. Justify.

4. Two functions, $f(x) = a|x|$ and $g(x) = |bx|$, look identical for every x . What relationship must a and b satisfy?

5. Factor $f(x) = 7|-2x + 8|$ into a $|x - c|$ form and give the real a-value and vertex x-coordinate.

Are $f(x) = -3|x|$ and $f(x) = 3|-x|$ the same graph? Justify algebraically, then describe every transformation present. Finally, factor $f(x) = 5|4x - 20|$ to find its real a-value.

Transformations of $f(x) = |x|$, Part II

Algebra II Honors | Unit 2 | Day 3 | TEK A2.6.C

Objective: I can derive why (c, d) is always the vertex of $f(x) = a|x - c| + d$, and work backward from a vertex and a point to find the equation.

Vocabulary

Horizontal shift — $f(x - c)$; shifts opposite the sign of c .

Vertical shift — $f(x) + d$; shifts the same direction as the sign of d .

Vertex form — $f(x) = a|x - c| + d$, written specifically to make the vertex _____ easy to read off.

Part 1 — Prove the Vertex Is Always (c, d)

Evaluate $f(x) = a|x - c| + d$ at $x = c$, for any a, c, d :

$$f(c) = a|c - c| + d = a|0| + d = a \cdot \underline{\hspace{1cm}} + d = \underline{\hspace{1cm}} \rightarrow \text{so } (c, d) \text{ is always ON the graph}$$

Explain why (c, d) is the LOWEST point when $a > 0$ (or highest when $a < 0$), not just A point on the graph.

Sample: _____

Part 2 — Working Backward: Vertex + One Point $\rightarrow$ Equation

A function has vertex $(3, -2)$ and passes through $(5, 4)$. Find a , then write the equation.

$$a|5 - 3| + (-2) = 4 \rightarrow a(2) - 2 = 4 \rightarrow 2a = \underline{\hspace{1cm}} \rightarrow a = \underline{\hspace{1cm}}$$

$$\text{Equation: } f(x) = \underline{\hspace{2cm}}$$

You Try. Vertex $(-1, 5)$, passes through $(2, -4)$. Find a and write the equation.

$$a|2 - (-1)| + 5 = -4 \rightarrow 3a = \underline{\hspace{1cm}} \rightarrow a = \underline{\hspace{1cm}} \quad f(x) = \underline{\hspace{2cm}}$$

A second one. Vertex $(4, 1)$, passes through $(7, 10)$. Find a and write the equation.

$$a|7 - 4| + 1 = 10 \rightarrow 3a = \underline{\hspace{1cm}} \rightarrow a = \underline{\hspace{1cm}} \quad f(x) = \underline{\hspace{2cm}}$$

Part 3 — Connecting Back to Day 2's Factoring

$f(x) = 2|3x - 12| + 5$ has a b-value hiding inside. Factor to find the vertex form $a|x - c| + d$.

$$2|3x - 12| + 5 = 2|3(x - 4)| + 5 = 2 \cdot \underline{\hspace{1cm}} \cdot |x - 4| + 5 = \underline{\hspace{1cm}}|x - 4| + 5 \rightarrow \text{vertex } \underline{\hspace{2cm}}$$

Part 4 — Real World: Dunhill Drafting's Roofline

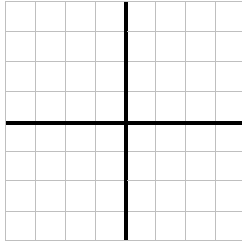
Roofline data was modeled by trial-and-error transformation, landing on $f(x) = -1.2|x - 3| + 8$.

Vertex (highest point of the roof): _____ Explain how you know it is a MAXIMUM, not a minimum, without graphing.

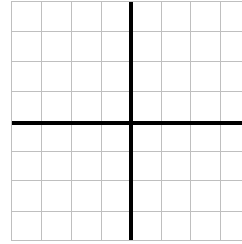
Part 5 — Graph and Compare

Graph both. Each square is 1 unit; both axes run -5 to 5.

A. $f(x) = |x - 2| - 3$



B. $f(x) = |x + 3| + 2$



Practice

1. Give the vertex of $f(x) = -3|x + 8| - 6$. _____
2. Factor $f(x) = 3|2x + 6| - 1$ into vertex form and give the vertex. _____ vertex

3. A function has vertex $(-2, 8)$ and passes through $(1, -1)$. Find a and write the equation.
 $a|1 - (-2)| + 8 = -1 \rightarrow 3a = \underline{\hspace{2cm}} \rightarrow a = \underline{\hspace{2cm}} \quad f(x) = \underline{\hspace{4cm}}$
4. Factor $f(x) = 5|10x - 30| + 2$ into vertex form, and give both the vertex and the real a -value.
_____ vertex _____ $a = \underline{\hspace{2cm}}$

Exit Ticket

A function has vertex $(2, -5)$ and passes through $(0, -1)$. Find a and write the equation, then explain how you know $(2, -5)$ is a minimum.

$a|0 - 2| + (-5) = -1 \rightarrow a = \underline{\hspace{2cm}} \quad f(x) = \underline{\hspace{4cm}}$

Name _____ Date _____ Period _____

Formulating Absolute Value Linear Equations

Algebra II Honors | Unit 2 | Day 4 | TEK A2.6.D

Objective: I can formulate an absolute value linear equation from a real-world description, including situations that need an extra structural step beyond the basic template.

Vocabulary

Formulate — set up an equation from a description without solving it.

Center value / margin — the basic template is $|x - \text{center}| = \text{margin}$ - but not every situation hands you those two numbers directly.

Part 1 — Pull Out Center and Margin Yourself

Clarissa wants a house within 5 houses of house #1248 on an even-numbered street.

Center: _____ Margin: _____ Equation: _____

A surveying instrument measures a 10-mile stretch with a 1.5% margin of error.

The 1.5% is a percent of the true distance, not a flat number of miles - it has to be built into the equation as a percent expression.

Equation: _____

Part 2 — A Layered Scenario: Brighton Builders' Hallway

A hallway has offices on both sides, evenly spaced from the center at 15x feet each way, and the total layout must equal 292 feet with 52 feet reserved for walls and doors.

This does not fit $|x - \text{center}| = \text{margin}$ directly - there is no subtraction from a center value. Identify what the absolute value expression represents here, and why.

Sample: _____

Equation: _____

Part 3 — Going Further: Generalize the Template

In your own words, explain what "center" and "margin" mean in a real-world $|x - \text{center}| = \text{margin}$ equation, and give one example of a situation where the equation does NOT fit that template (like Part 2).

Sample: _____

Part 4 — A Second Layered Scenario

A garden bed is built symmetric about a center walkway, with plant rows evenly spaced 4x feet from the walkway on each side, and the total bed width (rows plus a 3-foot walkway) is 27 feet.

Formulate the equation, and explain why it needs the layered approach rather than the basic template.

Equation: _____

Part 5 — Classify Before You Formulate

For each scenario, decide: does it fit the basic $|x - \text{center}| = \text{margin}$ template, or does it need the layered approach?

Scenario	Template or Layered?
A part must weigh 12 oz, +/- 0.5 oz	_____
A bridge has supports evenly spaced from its center, total span 200 ft	_____
A recipe needs 3 cups of sugar, within 1/4 cup	_____
A picture frame border is the same width on all sides, total outer width 24 in.	_____

Practice — Formulate, Do Not Solve

1. A shipping company's box-weight target is 50 lb, with a tolerance of 3 lb. _____
2. A factory's bolt length must be within 0.02 cm of 6 cm. _____
3. A rectangular sign is built symmetric about its center line, with total width 40 in. and each side extending 2x inches from center. Formulate, and explain why this is like Part 2, not Part 1.

Equation: _____

4. A quality-control spec allows a part's diameter to differ from 3 cm by at most 0.1 cm. Formulate and classify.
_____ Template

5. A fence is built symmetric about a center gate, with 6x feet of fencing on each side, and the total length including a 4-foot gate is 40 feet. Formulate and classify.

Equation: _____

Exit Ticket

A thermostat targets 70°F with an acceptable range of 2°F either way. Formulate the equation, then state whether it fits the basic template or needs the Part 2 approach, and why.

Equation: _____

Solving Absolute Value Linear Equations

Algebra II Honors | Unit 2 | Day 5 | TEK A2.6.E

Objective: I can justify why an absolute value equation splits into two cases, and solve absolute value linear equations that require multiple isolation steps.

Vocabulary

Isolate — get the absolute value expression alone before splitting.

Split into two cases — $|A| = k$ ($k > 0$) becomes $A = k$ OR $A = -k$.

Part 1 — WHY Does $|A| = k$ Split Into Two Cases?

Using Day 1's piecewise definition of absolute value, explain why $|A| = k$ ($k > 0$) means $A = k$ OR $A = -k$.

Sample: _____

Part 2 — Isolate First, Then Split

Example 1. $2|x + 1| - 3 = 7$

Isolate: $|x + 1| = \underline{\hspace{2cm}}$ Solve: $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

Example 2. $5 - 2|x - 1| = -3$

Isolate: $-2|x - 1| = \underline{\hspace{2cm}}$ $\rightarrow |x - 1| = \underline{\hspace{2cm}}$ Solve: $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

Notice: dividing by a NEGATIVE number does not flip anything here, unlike an inequality - it just isolates the absolute value.

Part 3 — Fractions and Distributing

Example 3. $|(2x - 4)/3| = 6$

$2x - 4 = 18$ or $2x - 4 = \underline{\hspace{2cm}}$ $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

Example 4. $|3(x - 2)| + 1 = 10$

Isolate: $|3(x - 2)| = \underline{\hspace{2cm}}$ $\rightarrow 3(x - 2) = 9$ or $3(x - 2) = \underline{\hspace{2cm}}$ $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

You Try. $|4(x + 1)| - 2 = 14$

$|4(x + 1)| = \underline{\hspace{2cm}}$ $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

Practice

1. $4|x + 2| = 20$ $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

2. $|(x + 5)/2| = 3$ $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

3. $8 - 3|x - 2| = -1$ $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

Exit Ticket

Solve $6 - 2|x + 3| = -4$, and explain in one sentence why the equation splits into two cases once $|x + 3|$ is isolated.

$|x + 3| = \underline{\hspace{2cm}}$ $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$ _____

Extraneous Solutions and Real-World Applications

Algebra II Honors | Unit 2 | Day 6 | TEK A2.6.D, A2.6.E

Objective: I can identify extraneous solutions - including cases where every solution is extraneous - and explain why checking in the original equation is necessary.

Vocabulary

Extraneous solution — a value the algebra produces that does NOT satisfy the ORIGINAL equation.

Part 1 — Why Checking the ORIGINAL Equation Matters

After splitting $|A| = k$ into $A = k$ or $A = -k$, both branches are true equations by construction. Explain why a solution can still fail once you plug it back into the ORIGINAL absolute value equation.

Sample: _____

Part 2 — A Solution That Does Not Check Out

Solve $3|x - 1| - 2 = 6x + 1$ and check BOTH answers.

$$|x - 1| = 2x + 1 \rightarrow x = \underline{\hspace{2cm}} \text{ or } x = \underline{\hspace{2cm}}$$

Check $x = -2$: $3|-3| - 2 = 7$, $6(-2) + 1 = -11$. _____ Check $x = 0$: $1 = 1$. _____

Part 3 — When EVERY Solution Is Extraneous

Solve $|x + 2| = -x - 5$ and check both.

Split: $x + 2 = -x - 5$ or $x + 2 = \underline{\hspace{2cm}}$

$x = \underline{\hspace{2cm}}$ or the second equation gives _____

Check $x = -3.5$: $|-1.5| = 1.5$, $-(-3.5) - 5 = -1.5$. $1.5 \neq -1.5 \rightarrow \underline{\hspace{2cm}}$

Solution set: _____

Why did this happen? Look at the right side of the original equation for a clue.

Sample: _____

Part 4 — Finish the Other Two Day 4 Equations

Clarissa's houses: $|x - 1248| = 5 \rightarrow x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$ Both are even-numbered addresses within 5 houses, so _____

Surveying: $1.5 = |x - 10|/10 \cdot 100 \rightarrow |x - 10| = \underline{\hspace{2cm}} \rightarrow x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$ miles

Part 5 — Finish Brighton Builders' Hallway

$2|15x| + 52 = 292 \rightarrow |15x| = \underline{\hspace{2cm}} \rightarrow x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$ (x is a suite count, so use $x = 8$)

That is 8 office suites per side per floor $\rightarrow$ _____ suites per floor $\rightarrow$ _____ total suites in the 20-story building

Why was $x = -8$ rejected here, even though it is not technically extraneous the way Part 2's answer was?

Part 6 — One More: Predict Before You Check

Solve $|2x + 5| = 3x + 1$. Before checking, predict: could BOTH solutions be valid, based on what the right side needs to be?

Prediction: _____

Split: $2x + 5 = 3x + 1$ or $2x + 5 =$ _____

$x =$ _____ or $x =$ _____

Check $x = 4$: $|13| = 13$, $3(4)+1 = 13$. _____ Check $x = -1.2$: $|2.6| = 2.6$, $3(-1.2)+1 = -2.6$.
_____ (matches the prediction - $-1.2 < -1/3$)

Practice

1. Solve $|2x - 1| = x + 4$ and check both. $x =$ _____ or $x =$ _____ (both valid)
2. Solve $|x - 6| = -x - 2$ and determine if any solution survives checking. _____
3. Before solving $|x + 10| = 2x - 3$, what must be true of x for any solution to be valid? _____
4. Solve $|x + 10| = 2x - 3$ and check. $x =$ _____ ($x = -7/3$ fails the $x \geq 1.5$ requirement, so it is rejected without even checking)
5. Solve $|4x - 1| = x + 8$ and check both. $x =$ _____ or $x =$ _____ (both valid, since $x + 8 \geq 0$ for both)
6. Solve $|5x + 2| = -3x - 5$. First state what x must satisfy for a solution to be possible, then solve, check, and give the solution set.

Requirement: _____ $x =$ _____ or $x =$ _____ - neither satisfies $x \leq -5/3$, so both are extraneous. Solution set: _____

Exit Ticket

Solve $|3x + 1| = x - 7$. Check both, and determine the solution set. Explain your reasoning for rejecting anything you reject.

Solving Absolute Value Linear Inequalities

Algebra II Honors | Unit 2 | Day 7 | TEK A2.6.F

Objective: I can justify why an absolute value inequality splits into an "and" or an "or" compound inequality, using the distance meaning of absolute value, and solve harder cases.

Vocabulary

Compound inequality — two inequalities joined by "and" or "or".

Part 1 — Why "And" and Why "Or"? Think Distance

Recall from Day 1: $|x|$ is the distance of x from 0. Use that to explain $|x| < k$.

$|x| < k$ means "x is within distance k of 0" - closer to 0 than k in EITHER direction, so x must be _____ -k and k. That is why $<$ gives an _____ statement.

Now explain $|x| > k$.

$|x| > k$ means "x is FARTHER from 0 than k" - either past k on the positive side, _____ past -k on the negative side. That is why $>$ gives an _____ statement.

Part 2 — Apply It

Example 1 ($<$, AND). $|2x + 3| < 5$

$$-5 < 2x + 3 < 5 \rightarrow \underline{\hspace{2cm}}$$

Example 2 ($\geq$, OR). $|2x - 3| + 4 \geq 9$

$$\text{Isolate: } |2x - 3| \geq 5 \rightarrow \underline{\hspace{2cm}}$$

Part 3 — Harder Cases

Example 3. $|3 - 2x| \leq 7$ (the "inside" is not in the usual $x - c$ order)

$$-7 \leq 3 - 2x \leq 7 \rightarrow -10 \leq -2x \leq 4 \rightarrow \text{dividing by } -2 \text{ flips both} \rightarrow \underline{\hspace{2cm}}$$

Example 4. $5|x + 1| - 3 > 12$

$$\text{Isolate: } |x + 1| > \underline{\hspace{1cm}} \rightarrow \underline{\hspace{2cm}}$$

You Try. $2|x - 4| + 1 \leq 9$

$$|x - 4| \leq \underline{\hspace{1cm}} \rightarrow \underline{\hspace{2cm}}$$

Practice

1. $|4 - x| > 6$ _____

2. $3|x - 2| < 12$ _____

3. Explain, using distance, why $|x + 2| \geq 0$ is true for EVERY real number x - no algebra needed.

Exit Ticket

Solve $|3x + 2| \geq 7$. Then, without solving, explain using distance why $|x - 5| < -2$ has no solution at all.

Name _____ Date _____ Period _____

Absolute Value Inequalities: Special Cases and Interval Notation

Algebra II Honors | Unit 2 | Day 8 | TEK A2.6.F

Objective: I can write absolute value inequality solutions in interval notation, and recognize both special cases: no solution, and all real numbers.

Vocabulary

No solution — an isolated absolute value set $<$ or $\leq$ a negative number.

All real numbers — an isolated absolute value set $>$ or $\geq$ a negative number - true no matter what x is.

Part 1 — Interval Notation Reference

Inequality	Compound form	Interval notation
$ x > 5$	_____	_____
$ x \geq 5$	_____	_____
$ x < 5$	_____	_____
$ x \leq 5$	_____	_____

Part 2 — Variable on Both Sides

Example. $|3x + 1| - 4 \leq 2x - 1$

Isolate: $|3x + 1| \leq 2x + 3 \rightarrow -(2x+3) \leq 3x+1 \leq 2x+3 \rightarrow$ _____

Part 3 — Two Special Cases, Side by Side

A. No solution. $4 + |x + 1| \leq 2$

Isolate: $|x + 1| \leq$ _____ An absolute value can never be _____. Solution: _____

B. All real numbers. $4 + |x + 1| \geq 2$

Isolate: $|x + 1| \geq$ _____ An absolute value is ALWAYS _____. Solution: _____

These two came from the SAME left side, just a flipped inequality symbol. In one sentence, explain why one has no solution and the other is always true.

Sample: _____

Practice

1. Solve $|x - 2| + 6 \leq 1$ _____

2. Solve $|x - 2| + 6 \geq 1$ _____

3. Solve $|2x - 1| > x + 5$ _____

Exit Ticket

Solve both $3 + |x - 4| < 1$ and $3 + |x - 4| > 1$. Explain in one sentence why they come out so differently despite looking almost identical.
